Rapid Review Template Protocol

Title	*Emerging Perspectives on GenAI for IoT Software Engineering: A Preliminary Review*
Context	Rapid technological advancement is enabling a wide range of applications and solutions. This has led to the rise of systems based on the Internet of Things (IoT) paradigm. These systems are composed of uniquely addressable objects (things) equipped with identification, sensing, and actuation capabilities, utilizing their built-in processing and communication to reach a goal [2, 3]. In addition, Generative Artificial Intelligence (*GenAI*) represents a more recent paradigm in technology, consisting of AI systems designed to generate text, images, and various other media via generative models [5]. The classification of this technology is primarily delineated by the training techniques employed in its development [6], as well as the specific architecture of the model being used [6]. The current paradigm in complex systems engineering, such as in smart factories marked by the intersection of the Internet of Things (IoT) and Generative Artificial Intelligence (GenAI). Many of these approaches leverage the versatility of IoT applications while utilizing the capabilities of GenAI to improve the efficiency of these systems, mitigate vulnerabilities, and automate processes[7]. Therefore, understanding the aspects of this type of system is necessary for the evolution of its development process, as well as to expose the pros and cons of this combination.
Motivation	The advancement of these technologies directly implies the need for the evolution of techniques and methods required for its production [3]. Consequently, there is a clear necessity for the continuous development of Software Engineering to support the construction of these contemporary systems with new techniques and approaches. However a robust definition of the most important features of this kind of system is necessary for the evolution of these techniques [1].
Problem	The introduction of systems that explore Generative Artificial Intelligence created an impact in a myriad of sectors, such as manufacturing and smart cities. This influence shows the potential that this kind of technology has to transform processes beyond acting as a support decision tool. Generally, the use of GenAI in the construction and idealization of these systems brings positive results in the literature. However, the implementation of this recent technology in software systems, mainly contemporary software systems such as IoT systems, also represents risks. Problems without solutions and lack of solid approach perspective, present in any new technology, allied with possible untraceable long term implications represents threats to the definition of solid characteristics. Consequently, characterizing these systems is essential for comprehending their features, architectures, and functionalities. This understanding is fundamental to conceiving this type of analysis in a structured way and to devising tools and approaches that guarantee the quality assurance for systems of this nature.
Goal	*Analyze GenAI in IoT software systems, for the purpose of characterization with respect to its use context, problem domains, models, possible drawbacks and advantages from the point of view of software engineering researchers in the context of IoT software systems projects utilizing models available in technical literature.*
Research Question	***RQ1:*** *How can GenAI be addressed in IoT software systems?* ***RQ2:*** *What problem domains of GenAI in IoT software systems are present in the technical literature?*

<table>
<tr><td></td><td>RQ3: What are the main challenges for using GenAI in IoT software systems?</td></tr>
<tr><td>Research questions for future work</td><td></td></tr>
<tr><td>Population</td><td>Technical Literature of Generative Artificial Intelligence in Internet of Things</td></tr>
<tr><td>Keywords</td><td>Generative Artificial Intelligence, Internet of Things, Rapid Review, Software Systems</td></tr>
<tr><td>Languages</td><td>English</td></tr>
<tr><td>Search Methods</td><td>● Automatic search in digital libraries
● Snowballing (one step forward and one step backward)</td></tr>
<tr><td>Sources</td><td>● Scopus (http://www.scopus.com/)</td></tr>
<tr><td>Alternatives Sources</td><td>● List relevant primary sources that were not found in the automatic search</td></tr>
<tr><td>Control Studies</td><td></td></tr>
<tr><td>Base search string</td><td>The research string employs the PICOC method [Petticrew & Roberts, 2006] for definition.
<table>
<tr><th>PICOC</th><th>STRING</th></tr>
<tr><td>Population</td><td>"IoT" OR "Internet Of Things" OR "Smart device*" OR "Connected device*" OR "Addressable thing" OR "Smart object*" OR "Smart ite*"</td></tr>
<tr><td>Intervention</td><td>"Generative AI" OR "Generative Artificial Intelligence" OR "Generative Model*" OR "Large Language Model*" OR "Language Model*" OR "Small Language Model*" OR "LLM*" OR "RAG" OR "Retrieval Augmented Generation" OR "Natural Language Processing" OR "NLP" OR "AI Agent" OR "LLM-based agent" OR "AI Multi-Agent"</td></tr>
<tr><td>Comparison</td><td></td></tr>
<tr><td>Outcome</td><td>"Use context" OR "Use case" OR "Advantage*" OR "Problem domain*" OR "Technique*" OR "Model*" OR "Research gap*" OR "Drawback*"</td></tr>
<tr><td>Context</td><td>“project” OR “product” OR “software system*”</td></tr>
<tr><td colspan="2">String

("IoT" OR "Internet Of Things" OR "Smart ite*" OR "Smart object*" OR "Addressable thing*" OR "Connected device*" OR "Smart device*") AND ("Generative AI" OR "Generative Artificial Intelligence" OR "Generative Model*" OR "Large Language Model*" OR "Language Model*" OR "Small Language Model*" OR "LLM*" OR "RAG" OR "Retrieval Augmented Generation" OR "Natural Language Processing" OR "NLP" OR "AI Agent" OR "LLM-based agent" OR "AI Multi-Agent") AND ("Use context" OR "Use case" OR "Advantage*" OR "Problem domain*" OR "Technique*" OR "Model*" OR "Research gap*" OR "Drawback*") AND (“project” OR “product” OR “software system*”)</td></tr>
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Scopus Search String	TITLE-ABS-KEY(("IoT" OR "Internet Of Things" OR "Smart ite*" OR "Smart object*" OR "Addressable thing*" OR "Connected device*" OR "Smart device*") AND ("Generative AI" OR "Generative Artificial Intelligence" OR "Generative Model*" OR "Large Language Model*" OR "Language Model*" OR "Small Language Model*" OR "LLM*" OR "RAG" OR "Retrieval Augmented Generation" OR "Natural Language Processing" OR "NLP" OR "AI Agent" OR "LLM-based agent" OR "AI Multi-Agent") AND ("Use context" OR "Use case" OR "Advantage*" OR "Problem domain*" OR "Technique*" OR "Model*" OR "Research gap*" OR "Drawback*") AND (“project” OR “product” OR “software system*”)) AND PUBYEAR > 2015 AND PUBYEAR < 2027 AND (LIMIT-TO (DOCTYPE,"cp") OR LIMIT-TO (DOCTYPE,"ar"))
Date of Research	*11/05/2026*
Inclusion Criteria (IC)	*IC1: The study presents evidence based on scientific empirical methods and technical reports (use context, techniques, projects and software systems) about GenAI in IoT Software Systems;* *IC2: The study answers at least one RQ;* *IC3: The study is a primary study.*
Exclusion Criteria (EC)	*EC1: The study does not meet at least one IC;* *EC2: The study is duplicated;* *EC3: The study is unavailable for free download or through institutional access;* *EC4: The study is not peer-reviewed or is a Master’s or Doctoral thesis;* *EC5: The study is an editorial, conference/workshop summary or report, table of contents, tutorial, keynote talk, or forward;* *EC6: The study is not written in English.*
Preliminary selection	Reading title and abstract of the retrieved studies.
Further selection	1. Reading the full text of selected studies.
Selection criteria	The criteria based on the step of selection are: ● Title: The population or intervention and related fields are cited in the title. A title is only excluded if it does not mention GenAI or IoT and related. ● Abstract: The abstract should meet at least one IC and should not meet any EC. Additionally, the abstract should mention at least one term of the PICOC fields but not as an example. Optimally, the abstract mentions an application of GenAI on IoT

Use of Generative AI	We acknowledge the use of AI generative tools to support text-processing tasks: translation, rewriting to improve clarity, condensing text to meet page limits, and drafting this acknowledgment. We reviewed all outputs to ensure alignment with our original intent.
References	[1] Laura Freeman, John Robert, and Heather Wojton, "The Impact of Generative AI on Test & Evaluation: Challenges and Opportunities," in *Proc. 33rd ACM Int. Conf. on the Foundations of Software Engineering (FSE Companion '25)*, 2025, pp. 1376–1380, doi: 10.1145/3696630.3728723. [2] Motta, R. C., V. Silva, and G. H. Travassos, "Towards a more in-depth understanding of the IoT Paradigm and its challenges," *Journal of Software Engineering Research and Development*, vol. 7, 3:1–3:16, 2019, doi:10.5753/jserd.2019.14. [3] Motta, Rebeca C., Káthia M. de Oliveira, and Guilherme H. Travassos, "An evidence-based roadmap for IoT software systems engineering," *Journal of Systems and Software*, vol. 201, 111680, 2023, doi:10.1016/j.jss.2023.111680. [4] Sengar, S.S., Hasan, A.B., Kumar, S. et al., "Generative artificial intelligence: a systematic review and applications," *Multimed Tools Appl*, vol. 84, pp. 23661–23700, 2025, doi: 10.1007/s11042-024-20016-1. [5] Schneider, J., "Explainable Generative AI (GenXAI): a survey, conceptualization, and research agenda," *Artif Intell Rev*, vol. 57, 289, 2024, doi: 10.1007/s10462-024-10916-x. [6] Shivani Pahuja, Sonal Kukreja, and Akansha Singh, "Comprehensive Review of Generative artificial Intelligence: Mechanisms, Models, and Applications," *Procedia Comput. Sci.*, vol. 258, C, 2025, pp. 3731–3740, doi: 10.1016/j.procs.2025.04.628. [7] Kar, Arpan & Ps, Varsha & Rajan, Shivakami. (2023). Unravelling the Impact of Generative Artificial Intelligence (GAI) in Industrial Applications: A Review of Scientific and Grey Literature. Global Journal of Flexible Systems Management. 24. 10.1007/s40171-023-00356-x. Diagnosis Using LLM," in *Proc. ICASSP, IEEE Int. Conf. on Acoustics, Speech and Signal Processing*, 2025, doi: 10.1109/ICASSP49660.2025.10888670.